

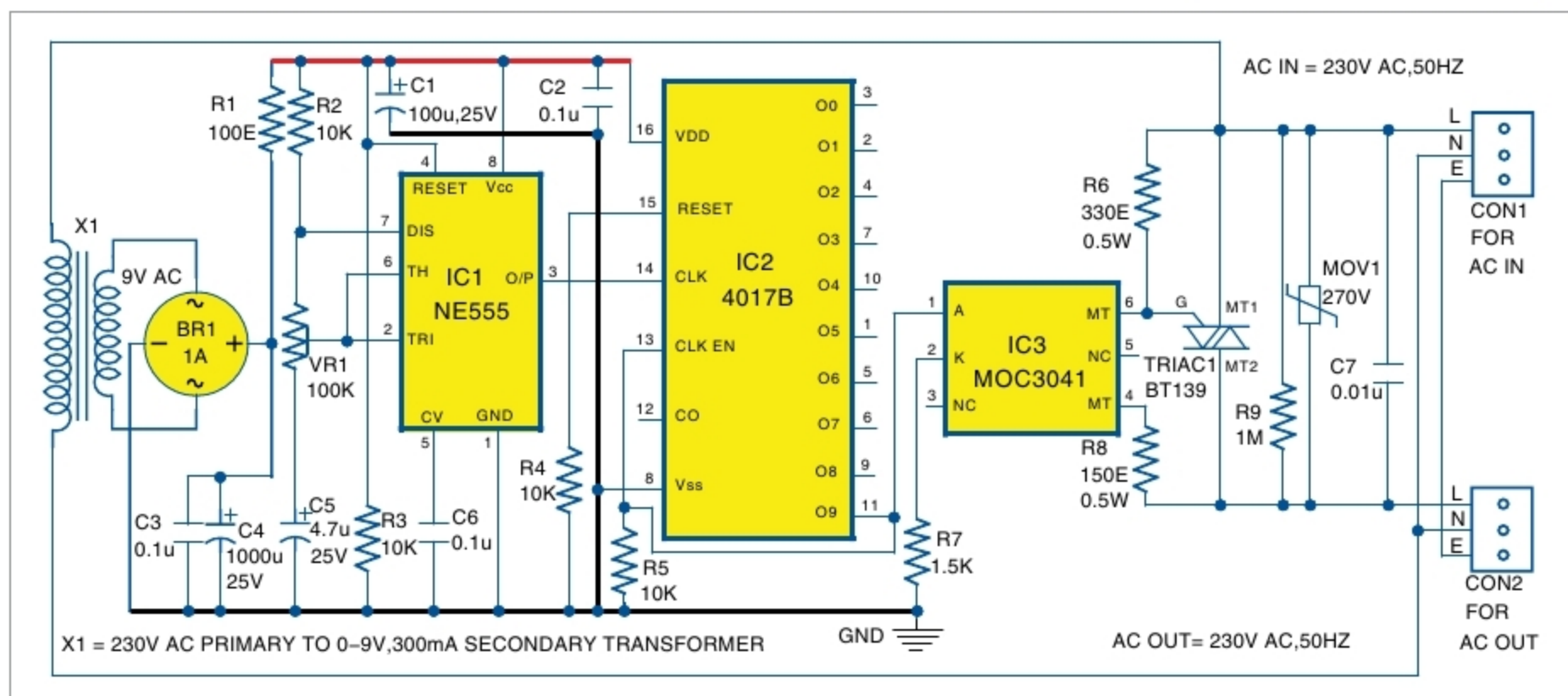


Fig. 3: Circuit diagram of the power-on delay timer

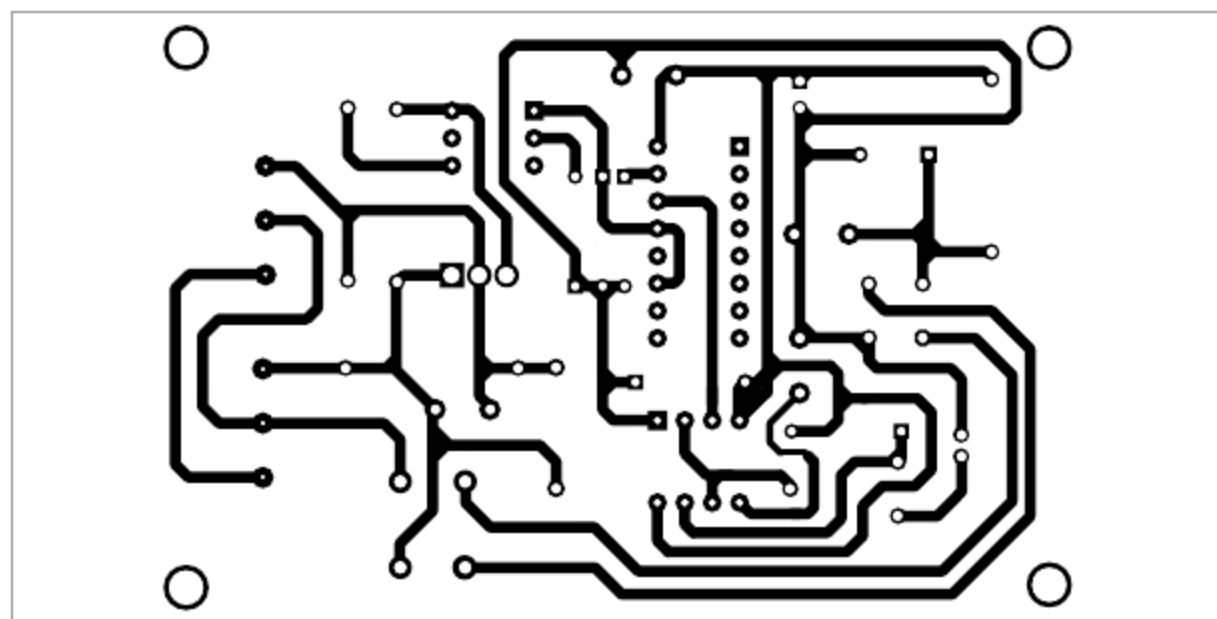


Fig. 4: Actual-size PCB layout for the power-on delay timer

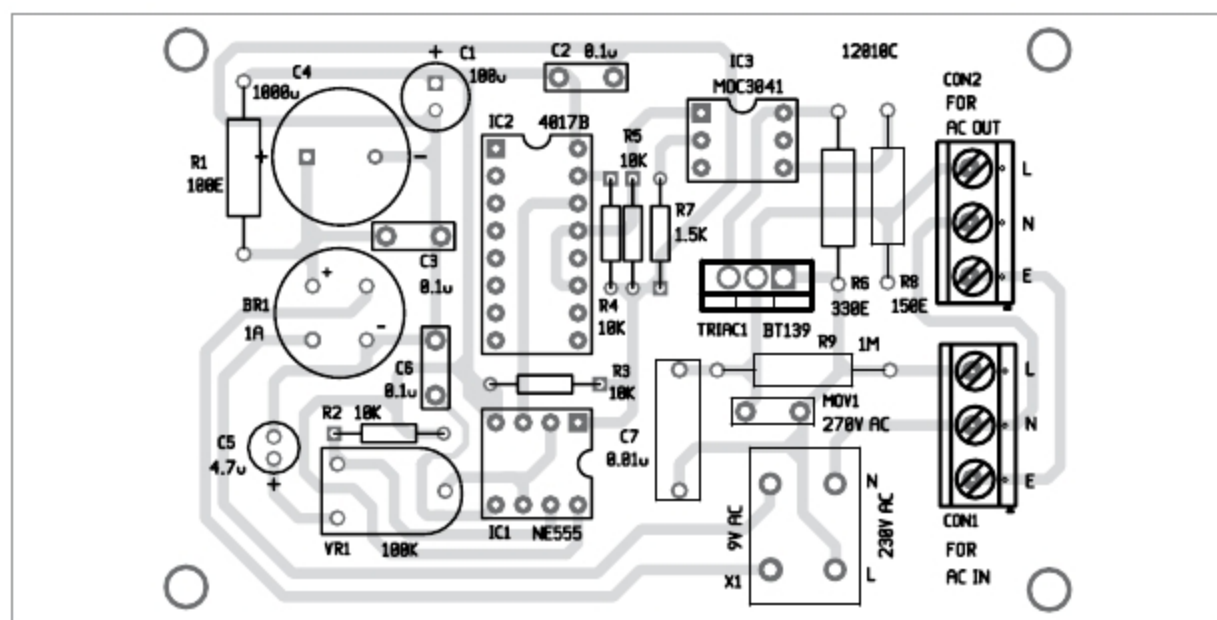


Fig. 5: Components layout of the PCB

4017B. The current high output states of 4017B remain high till the power is off. High state at Q9 makes MOC3041 to conduct. This triggers the main triac

BT139 and the AC supply for the load becomes available across connector CON2.

To improve the life of triac, the

spike arrester circuit built around resistor R9, metal oxide varistor MOV1 and capacitor C7 is used.

Construction and testing

An actual-size PCB layout for the power-on delay timer is shown in Fig. 4 and its components layout in Fig. 5. After assembling the circuit, connect the 230V, 50Hz AC supply across CON1. AC load should be connected across CON2.

For initial testing, remove IC3 from the circuit and connect an LED across pins 1 and 2 of MOC3041. Switch on the delay timer circuit. The LED should glow after a delay. If LED does not glow, check voltage at pin 3 of NE555 with multimeter. If voltage pulses are available at pin 3 of IC NE555, check 4017B and its associated components.

If LED glows after a time delay, your circuit is working well. Now, remove the LED and insert the opto-triac MOC3041, triac BT139 and associated components. The power-on delay timer is now ready for use.

A suitable heat-sink is recommended for triac BT139. You may use BTA12 in place of BT139. **EFY**

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